

Adrian Palamarev

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA

School of Computer Science: Bachelor of Science in Robotics and Computer Science

May 2029

Selected Coursework: Computer Systems, Parallel and Sequential Data Structures and Algorithms, Functional Programming, Imperative Programming, Mathematical Foundations for Computer Science

GPA: 4.0

University of Florida, Gainesville, FL

August 2023 – May 2025

Selected Dual-Enrollment Coursework: Programming Fundamentals 1 and 2, Differential Equations, Calculus III – GPA: 4.0

SKILLS

Strengths: Algorithm development, parallel programming, data structure design, numerical methods, databases, user interfaces

Programming Languages: C++, C#, C, Python, SML, Rust, Java, HTML, CSS, JavaScript, WASM, SQL, Typst, LaTeX

Frameworks: CUDA, SIMD, ROS2, PyTorch, .Net

SELECTED PROJECTS

High-Performance Hash Table | C++, SIMD

June 2026 – August 2026

- Created a highly optimized, memory-efficient flat hash table in C++ using variable linear probing and SIMD operations. Achieved comparable insertion and lookup performance to tables using Google's SwissTable layout.

Parallel-Friendly Memory Allocator | C++

May 2026 – July 2026

- Designed and implemented a highly efficient memory allocator, leveraging paged data structures. Achieved strong scaling, and exceeded the performance of Microsoft's mimalloc on workloads with large active memory footprints. Achieved a speedup of 1.44 times over Microsoft's mimalloc in a high-churn, 10 million operation test.

Web Assembly Teaching Game (HackCMU) | Rust, WASM

September 2025

- Worked in a team to create an interactive and educational game that teaches learners Web Assembly Programming through several levels. Advanced my teamwork skills and strengthened my understanding of Web Assembly code.

Radio Propagation Plot Digitalization Software | C#, .NET

July 2025 – August 2025

- Developed image processing algorithms in C# to digitize highly distorted plots using convolution and Fourier filtering, reducing processing time by over 90 percent.
- Contributed to creating the user interface for processing the plots.

Math, Statistics, and Neural Network Libraries | C++, CUDA

July 2021 – July 2025

- Aggregated my mathematics and neural network knowledge into libraries and strengthened my C++ proficiency; wrote over 30,000 lines of code. Designed and implemented a parser and interpreter for a simple mathematical language.
- Developed a neural network from scratch in C++ and trained it using matrix-based gradient descent algorithms; gained substantial knowledge in GPU (CUDA) and CPU parallel programming, and created a GPU CUDA optimized Matrix and Linear Algebra library.

Finance Tracker – Future Business Leaders of America | C#, .NET, SQL

November 2024 – March 2025

- Implemented a full-stack, client-server application for tracking incomes and expenditures across categories utilizing the C# and the .NET framework and an SQL database to securely store transactions and user information.
- Won first place at the state convention and acquired a much better understanding of working with SQL database functionality.

EXPERIENCE

Peer Tutor – Carnegie Mellon Student Academic Success Center

August 2026 – Present

- Ran structured one-hour tutoring sessions for students in Computer Systems, Imperative Programming, and Discrete Math classes
- Created supplemental materials, explained challenging questions, and shared efficient problem solving and study strategies

SELECTED CLUBS

Carnegie Mellon Robotics Club

Member 2025 – Present

Carnegie Mellon Cycling Club

Member 2026 – Present

SELECTED AWARDS & PLACEMENTS

FBLA Business Calculations – 2nd Place at the National Convention

July 2024

High School Mathematical Contest in Modeling (A Cost-Benefit Analysis of Electric Buses) – Finalist Status

November 2023

USA Computing Olympiad Silver Division

January 2023